Eid-ul-Adha Vacation
Practice Problem Set

📋 **How to Use This Problem Set**

- Every question has both a Theory part and a Practical (coding) part — you must attempt both.
- Questions marked 🔬 RESEARCH require you to read about the dataset domain before answering.
- You may use Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn — same libraries used in previous class.
- Surprise bonus challenge at the end — don't peek until you finish the main questions!

Q1 The Income Trap

DATASET	Adult Income (Census Income) https://archive.ics.uci.edu/dataset/2/adult
RESEARCH	Read about what 'capital-gain' and 'capital-loss' represent in US income tax filings before answering.

Theory

1. The 'capital-gain' column has ~91% of its values equal to exactly zero, with a few extreme values reaching over $99,000. Is this an outlier problem or a distributional problem? Justify your answer using the concept of the Normal Distribution from class.
2. If you apply StandardScaler (Z-score normalization) to 'capital-gain', what happens to the Z-scores of those extreme values? Why would a RobustScaler be more appropriate here?

3. Explain Winsorization. If you apply 5th–95th percentile Winsorization to 'hours-per-week', what are you assuming about the data-generating process, and what are the risks?

Practical

4. Load the dataset (it uses '?' for missing values — handle this first). Perform train-test split. Plot the distribution of 'capital-gain' using a KDE plot. Confirm the extreme skew visually.
5. Apply Winsorization (clip at 5th–95th percentile) to 'capital-gain' on the training set. Apply the SAME clipping bounds to the test set. Plot before/after KDE plots.
6. Apply RobustScaler to 'hours-per-week' and 'capital-gain' . Compare the scaled range with what StandardScaler would have produced. Which handles the outliers better? Justify numerically.

Q2 The Churning Machine

📦 DATASET	Telco Customer Churn https://www.kaggle.com/datasets/blastchar/telco-customer-churn
🔬 RESEARCH	Research what 'churn' means in the telecom industry and why it is a critical business KPI before starting.

Theory

7. The target column 'Churn' has roughly 73% 'No' and 27% 'Yes'. Explain why this class imbalance matters for a model. If you don't address it, what kind of predictions might the model make?
8. The 'TotalCharges' column is stored as a string (object) despite being numeric. What is the likely reason? (Hint: look at what happens with customers who have 0 tenure.) What preprocessing step is needed?
9. From an EDA perspective, what is the difference between a countplot with hue and a grouped bar chart (barplot)? When would you use each for this dataset?

Practical

10. Load the dataset. Fix the 'TotalCharges' data type issue. Confirm by checking df.info() before and after.

11. Perform full univariate analysis: (a) For all categorical columns, create countplots; (b) For all numerical columns, plot histograms with kde=True. Write at least one business insight per plot.
12. Perform multivariate analysis: (a) Create a countplot of 'Contract' vs 'Churn' with hue; (b) Create a KDE plot of 'tenure' grouped by 'Churn';Describe two findings that would help a business reduce churn.

Q3 Z-Scores & The Empirical Rule in the Wild

DATASET	Medical Cost Personal Dataset https://www.kaggle.com/datasets/mirichoi0218/insurance
RESEARCH	Research what BMI (Body Mass Index) is, what the clinical thresholds ar (underweight, normal, overweight, obese), and why it might be a critical feature for insurance charges.

Theory

13. According to the Empirical Rule (68-95-99.7), if BMI is approximately normally distributed with mean=30 and std=6, what percentage of people have BMI above 42?
14. You detect that the 'charges' column has a Z-score outlier threshold at $|Z| > 3$. A student removes ALL rows with $|Z| > 3$ before the train-test split. Identify two problems with this approach.
15. Why might the IQR method detect different outliers than the Z-score method on the same column? In which situation is each method preferred?

Practical

16. Load the dataset. Verify whether 'bmi' follows an approximately normal distribution: (a) Plot a KDE plot; (b) Compute skewness
17. For the 'charges' column: (a) Identify outliers using the IQR method (b) Identify outliers using the Z-score method (c) Compare how many outliers each method detects. Which is more conservative?
18. Create a new feature 'bmi_category' using the WHO thresholds (Underweight < 18.5, Normal 18.5–24.9, Overweight 25–29.9, Obese >= 30). Then use a barplot to visualize average 'charges' per BMI category. What pattern do you see?

Q4 The Big City Problem — Outlier Handling at Scale

DATASET	NYC Taxi Trip Duration https://www.kaggle.com/c/nyc-taxi-trip-duration use train.csv, with 10 or 20k data
RESEARCH	Research the geography of New York City's boroughs. Understand the maximum realistic trip duration and distance in NYC. This is essential to define what constitutes a real outlier.

Theory

19. A trip duration of 86,400 seconds (24 hours) exists in this dataset. Is this a data entry error, a real event, or something else? How would you decide programmatically AND using domain knowledge?
20. After removing outliers from the training set, should you also remove outliers from the test set? Justify your answer carefully — think about what a test set represents in a real production scenario.

Practical

21. Load the dataset (sample 20k rows). Convert 'pickup_datetime' to datetime and extract hour, day-of-week, and month as new features. Check: are there any biologically impossible trip durations (e.g., <= 0 seconds or > 24 hours)?
22. Apply IQR-based outlier detection to 'trip_duration'. Print how many outliers were detected and what % of the data they represent. Remove them from the training set ONLY.
23. Create a scatter plot of pickup_longitude vs pickup_latitude, coloring points by trip_duration. Do the outlier trips cluster in any particular geographic area? What does that suggest?

Q5 Happiness by the Numbers — Full EDA Story

DATASET	World Happiness Report 2023 | Kaggle: https://www.kaggle.com/datasets/ajaypalsinghlo/world-happiness-report-2021
RESEARCH	Read the executive summary of the World Happiness Report methodology (available on worldhappiness.report). Understand what the 'Ladder score', 'Logged GDP', and 'Social support' variables actually measure.

🧠 Theory

24. The 'Logged GDP per capita' column already has a log transformation applied. Why might the original GDP per capita have been log-transformed before inclusion in this report? What distribution property does log-transformation fix? **(Optional : Haven't covered in the module , you can study it by yourself. We will cover it if it seems necessary)**

25. You are asked to predict a country's happiness score. You have 'Social support', 'Healthy life expectancy', and 'Freedom to make life choices' as features. Explain conceptually whether you expect these will impact a man happiness score
26. A student concludes from a scatterplot that 'high GDP causes high happiness'. Critique this conclusion. Explain your thoughts

🧪 Practical

27. Load the dataset. Identify the top 10 and bottom 10 happiest countries. Create a horizontal bar chart showing both groups side-by-side with different colors.
28. Create a scatterplot of 'Logged GDP per capita' vs 'Ladder score'. Add a regression line using sns.regplot().
29. Perform regional analysis: group by 'Regional indicator' and compute the mean of all numeric columns. Visualize mean happiness by region using a barplot sorted by score. Which region has the highest variance in happiness? Use a boxplot to verify.

SURPRISE BONUS CHALLENGE

The End-to-End Pipeline Challenge

Pick ANY one of the 5 datasets above. Build a complete, end-to-end sklearn ColumnTransformer that:

- Handles missing values with the correct strategy for each column type
- Encodes all categorical features appropriately for the model
- Applies the right scaler (Standard vs. Robust — justify your choice)
- Handles outliers using either Zscore , IQR clipping or Winsorization (before the pipeline)



Eid Mubarak! 🌙 Happy Learning — See you after the vacation!